

Construction of 30m long PSC Girder Bridge Over Karela River
Upazila: Pirganj, District: Rangpur

Design Load = 3750 kN

Grip Length = ~~597~~ 597 mm

Jack ID: L804019

Pump ID: 803189

Gauge ID: 2409120

$$Y = 87.724X - 37.955$$

$$X = \frac{Y + 37.955}{87.724}$$

$$X_{20} = 8.98$$

$$X_{40} = 17.53$$

$$X_{60} = 26.08$$

$$X_{80} = 34.63$$

$$X_{95} = 41.04$$

$$X_{98} = 42.33$$

$$X_{100} = 43.18$$

$$X_{102} = 44.04$$

$$X_{105} = 45.32$$

Jack ID: L804022

Pump ID: 803180

Gauge ID: K-01 (18.12.213)

$$Y = 89.717X - 58.613$$

$$X = \frac{Y + 58.613}{89.717}$$

$$X_{20} = 9.01$$

$$X_{40} = 17.37$$

$$X_{60} = 25.73$$

$$X_{80} = 34.09$$

$$X_{95} = 40.36$$

$$X_{98} = 41.62$$

$$X_{100} = 42.45$$

$$X_{102} = 43.29$$

$$X_{105} = 44.54$$

Design Data:

Area of strand = 140 mm²

Area of cable, $A_1 = 2660$ mm²

Modulus of Elasticity of cable $E_1 = 197000$ MPa

Avg slip = 6 mm

Test Data:

Area of strand = 142.50 mm²

Area of cable $A_2 = 2707.50$ mm²

Modulus of Elasticity of cable $E_2 = 200400$ MPa

Elongation for Grip length:

$$\delta = \frac{PL}{AE}$$

$$= \frac{3750 \times 1000 \times 597}{2660 \times 19700}$$

$$= 4.27$$

Correction factor:

$$\frac{A_1 E_1}{A_2 E_2} = \frac{2660 \times 197000}{27075 \times 200400}$$

$$= 0.966$$

Cable No	Design			Correction factor	Corrected Elongation	Corrected Elongation for Grip
	Elongation	Elongation for Grip length	Total Elongation			
3	150 mm	4.27 mm	154.27 mm	0.966	149.02	4.12 mm